CUSTOM ANALOGUE DJ / DUB MIXER

Project status, engineering summary, and current phono-input design - 18 September 2026

Purpose: consolidated reference of the project as currently defined, separating locked decisions, proposed engineering targets, open decisions, and superseded ideas.

Important: this remains a design-status document, not a production-ready schematic, PCB or BOM. Values marked proposed require simulation and measurement after topology selection.

Status labels

- **LOCKED** - explicitly established functional/design decision.
- **PROPOSED** - useful starting value or engineering target; not frozen.
- **OPEN** - requires technical or creative decision.
- **SUPERSEDED** - earlier idea replaced by the current architecture.

1. Executive status

The overall system is technically feasible as a high-end custom analogue DJ / dub mixer. No major subsystem identified so far is fundamentally impossible. The project is now moving from architecture into actual circuit engineering.

The main difficult blocks are the global state-variable filter, resonance/self-oscillation, the five-way KILL/filter bank, five-band creative EQ, analogue CV distribution, the complete noise budget, power/ground architecture, and the modular backplane.

Engineering method from this point: requirement -> transfer function -> topology -> component values -> simulation -> PCB/layout -> prototype -> measurement -> revision.

Feasibility snapshot

Area	Status
Overall analogue concept	LOCKED / feasible
4 DJ channels	LOCKED / feasible
Phono/RIAA	Function locked; circuit now in Rev A design
DJ 4-band EQ	Function locked; topology open
Global filter	Function locked; technically demanding
5-way isolator/EQ	Function locked; major circuit task
4 FX buses + cross-feedback	LOCKED / feasible
Mic + generic mono	Function locked; topology open
Master/booth/record	Architecture locked; output topology open
Modular backplane	Concept locked; implementation open
Noise / bandwidth targets	Proposed; require validation

2. Global locked principles

- **100% analogue audio.** No ADC, DAC, DSP or digital audio processing.
- **Stereo master L/R.** Mono sound-system operation is downstream.
- **No pan or separate L/R user controls.** Stereo controls are linked.
- **Modular/serviceable construction.** Functional cards/modules use defined interfaces; not necessarily literal Eurorack electrical standard.
- **No hidden digital audio control.**
- **Extreme settings allowed** for filtering, saturation and feedback, while electrical operation remains non-destructive.
- **No intentional master compressor/limiter/clipping.** Electrical protection is still required.
- **No hot swapping.**

3. Master architecture

Stage	Signal flow
DJ CH1-4	-> DJ BUS -> ONE 5-WAY -> MASTER SUM
FX returns	-> MASTER SUM; post-5-way only
MIC	-> MASTER SUM

Stage	Signal flow
GENERIC MONO	-> MASTER SUM
MASTER	MASTER SUM -> MASTER LEVEL -> MASTER METER -> MASTER OUT
BOOTH	Independent level -> BOOTH L/R
REC	Fixed pre-master-level feed -> REC L/R

Outputs

Output	Specification
MASTER	Balanced XLR; -infinity -> +6 dB; main DSP feed
BOOTH	L/R; independent -infinity -> 0 dB
REC	Fixed pre-master-level; about 0 dBu nominal
Driver	Active balanced low-Z; ideally <50 ohm
Headroom	Target >= +24 dBu clean where practical
Protection	Current/short-circuit, DC/RF/transient as appropriate; power mute
Dynamics	No limiter/compressor

4. Global gain and headroom reference

Level	Vrms	Vpk	Vpp	Use
0 dBu	0.775	1.096	2.19	Nominal internal reference
+4 dBu	1.228	1.736	3.47	Professional line reference
+10 dBu	2.451	3.466	6.93	Hot programme
+18 dBu	6.155	8.704	17.41	Extreme internal level
+24 dBu	12.283	17.37	34.74	High-headroom target

- Normal programme around 0 dBu; +10 dBu hot; +18 dBu extreme.
- Critical stages: proposed >= +24 dBu clean where practical.
- **Correction:** +/-24 V may be the main system rail, but is not a universal op-amp supply. Local +/-15 V or +/-12 V rails may be required.
- **Correction:** +24 dBu is a stage/system headroom target, not a requirement that every IC itself handle that level.

5. DJ channels - four identical stereo channels

Signal flow: stereo RCA -> protection -> PHONO/LINE architecture -> GAIN -> 4-band EQ -> split to CUE / FX SEND / filter path -> meter -> fader -> DJ BUS.

Architecture correction introduced by the phono design: PHONO and LINE should be treated as two dedicated input paths that are selected after their respective front ends, rather than one switch being assumed to precede both RIAA and line buffering.

Function	Current specification
Input	Stereo RCA; PHONO/LINE
Ground	Dedicated turntable ground
Gain	-infinity -> 0 -> +10 dB; centre detent 0
Gain tracking	Stereo linked; target <=0.5 dB L/R
EQ	4 bands; -infinity -> 0 -> +6 dB
EQ tracking	Target <=0.25 dB L/R
CUE	Post-EQ, pre-filter, pre-fader
Fader	DJ-style conductive plastic; 0 dB max -> -infinity
Fader buffer	Target <=50 ohm

DJ EQ - proposed values

Band	Centre	Territory	Q
LOW	70 Hz	20-150 Hz	~0.7
LOW-MID	250 Hz	100-600 Hz	~0.8
MID-HIGH	1 kHz	400 Hz-2.5 kHz	~0.9
HIGH	5 kHz	1.5-20 kHz	~0.7

- Broad musical DJ correction/matching; not sound-system band pushing.

- 0 = flat/unity. Current topology concept: reference + controlled band contribution paths.
- Control law: faster action near extreme cut/boost; slower near the useful centre.

6. NEW ENGINEERING STEP - PHONO INPUT DESIGN (Rev A)

Scope: one identical stereo MM phono front end per DJ channel, feeding the common DJ gain stage. Rev A is intentionally MM-focused; MC is not included unless later requested.

Recommended architecture

Block	Recommended Rev A design
Connector	Stereo RCA L/R + dedicated turntable ground
Input protection	100 ohm series resistor per leg; low-capacitance protection physically after the resistor; avoid large TVS capacitance directly across cartridge
MM load	47.0 k ohm to signal ground, 0.1%
MM capacitance	100 pF C0G/NP0 default; provision for internal 0 / 47 / 100 / 150 pF selection so cable capacitance can be included
First gain stage	OPA1654 JFET-input audio op-amp; one stereo DJ channel uses four amplifier sections (two per L/R path); high-Z cartridge interface
First-stage gain	Target 21.3 dB (gain = 11.615); starting resistors 17.0 k ohm / 1.60 k ohm in a non-inverting gain cell
RIAA topology	Hybrid active/passive approach: first gain stage -> passive 75 us section -> second gain stage implementing remaining RIAA time constants
Second gain stage	OPA1654; active RIAA network implementing 3180 us and 318 us portions; total 1 kHz gain target initially 40-42 dB
DC strategy	Rev A candidate: low-frequency AC coupling between stages with sufficiently large low-leakage capacitor and 100 k ohm bias; direct-coupled servo remains a Rev B option
RIAA accuracy	Target <= +/-0.5 dB; preferred <= +/-0.25 dB from 20 Hz to 20 kHz after simulation/measurement
Output drive	Low-Z buffer into PHONO/LINE selector and DJ gain; target >= +18 dBu clean from the phono block
Local supply	Prefer regulated +/-15 V analogue rails for the phono module, with strong local decoupling and physical separation from control/PSU noise

Why OPA1654 for the phono front end

The MM cartridge is a high-impedance, inductive source. A FET-input audio amplifier is attractive at the cartridge interface because input current noise is extremely low. TI specifies the OPA1652/1654 family at about 4.5 nV/sqrt(Hz) voltage noise at 1 kHz, 3 fA/sqrt(Hz) current noise, 18 MHz GBW, unity-gain stability and operation up to +/-18 V. That makes the family a credible MM front-end choice. The OPA1612 remains a strong candidate for lower-impedance downstream stages, but its bipolar input makes it less attractive as the first MM device when cartridge impedance rises with frequency.

RIAA topology choice

For this mixer I recommend the two-stage hybrid active/passive RIAA approach rather than putting the complete equalization into one aggressive feedback loop. A real, measured MM implementation from audioXpress uses a JFET-input OPA1642, a first-stage gain of 21.3 dB, a passive 75 us section, a second stage implementing the 318 us and 3180 us constants, and about 42 dB total gain at 1 kHz. Its reported measurement showed about +/-0.1 dB RIAA deviation from 20 Hz to 10 kHz. A Texas Instruments application report likewise documents passive and active/passive RIAA topologies and emphasizes gain distribution for noise versus overload. These are references for topology, not a claim that their exact values are already final for this mixer.

RIAA component policy

- Use 0.1% metal-film resistors in the RIAA network.
- Use low-DA film capacitors, preferably polypropylene, for the RIAA time-constant network; C0G/NP0 is appropriate for small-value RF/load capacitors.
- Keep the 47 k ohm cartridge load physically at the RCA input connector.
- Keep the selectable cartridge capacitance physically close to the input connector.
- Keep the phono input area physically away from PSU switching nodes, control CVs, LEDs and headphone circuitry.
- Use a defined local phono signal-ground reference and a controlled chassis/earth connection rather than relying on arbitrary cable shields.
- No intentional 8-15 Hz rumble filter in Rev A; preserve the specified RIAA transfer and evaluate rumble filtering later as an optional function.

Preliminary electrical targets for the phono block

Target	Rev A starting value
Cartridge type	MM only
Load resistance	47.0 k ohm
On-board load capacitance	100 pF default; selectable alternatives internally
First-stage gain	21.3 dB

Target	Rev A starting value
Total gain at 1 kHz	40-42 dB target
Overload target	Aim for >=20 dB above nominal 5 mV @ 1 kHz cartridge signal at the input, subject to final gain distribution
Phono output nominal	Designed to be in the sub-line-level region so the DJ GAIN stage can normalize it
Bandwidth	Full RIAA 20 Hz-20 kHz; no intentional LF truncation in Rev A
Noise	First-stage noise is the priority; final module noise to be budgeted separately
Verification	Inverse-RIAA measurement + 1 kHz gain + overload + noise + channel tracking + crosstalk

What is still needed from you for the phono design

- **1. Cartridge scope:** Is V1 definitely MM-only? If you want high-output MC or low-output MC, the front end changes materially.
- **2. Cartridge loading:** Do you want a fixed 100 pF on-board capacitance, or internal selectable 0/47/100/150 pF? My recommendation is selectable internally, with 100 pF as the default.
- **3. Gain target:** Should I freeze 40 dB or 42 dB at 1 kHz? I currently favour about 42 dB because it works well with common 5-8 mV DJ cartridges while leaving the following DJ GAIN stage useful.
- **4. User access:** Should cartridge loading be technician/solder-jumper only, or do you want a rear-panel/user-accessible selector?

Current recommendation pending your answers: MM-only, 47 k ohm load, 100 pF default with internal alternatives, OPA1654 JFET-input front end, 21.3 dB first stage, hybrid RIAA, 40-42 dB total at 1 kHz, and no intentional rumble filter in Rev A.

7. Global DJ filter system

Locked correction: frequency, resonance and overdrive are shared controls; each DJ channel only selects HPF / OFF / LPF.

- HPF: 20 Hz -> 1.5 kHz.
- LPF: 500 Hz -> 20 kHz.
- 24 dB/oct target.
- Candidate: 4-pole state-variable architecture, conceptually two cascaded 2-pole sections.
- 0-5 V is only a proposed control interface convention.
- Resonance: 0-70% normal, 70-90% aggressive, 90-100% self-oscillation.
- Proposed oscillation level: about +6 to +10 dBu, softly bounded.
- Overdrive follows filter/resonance and precedes the DJ meter; JFET then asymmetric-BJT prototypes.
- Filter mode switching: break-before-make with about 1-5 ms click suppression.

8. Five-way sound-system / isolator stage

DJ BUS -> ONE 5-WAY -> MASTER SUM. FX returns bypass the 5-way. External DSP performs the acoustic crossover.

- Physical KILL map: SUB ~20-90 Hz; KICK ~90-300/400 Hz; MID ~300/400 Hz-1.2 kHz; HIGH ~1.2-3.5 kHz; SUPER HI ~3.5-20 kHz.
- Creative ranges: 20-120 Hz, 80-400 Hz, 250 Hz-1.5 kHz, 800 Hz-5 kHz, 3-20 kHz.
- Max boosts: +15, +15, +15, +12, +9 dB respectively.
- Width target: narrow ~Q3, medium ~Q1, broad ~Q0.5.
- KILL must be designed as a coherent complementary filter bank and positioned so creative EQ cannot restore killed material.
- Five-way drive follows the EQ sum and precedes the 5-way meter.

9. FX system

Element	Current specification
Buses	Exactly 4
Sources	DJ, mic, generic mono, FX returns
Send point	Post-EQ / pre-filter
Send level	-infinity -> 0 dB
Send EQ	LOW ~120 Hz / HIGH ~4.5 kHz; -infinity -> 0 -> +6 dB
Return gain	-infinity -> 0 -> +20 dB
Return EQ	LOW ~100-150 Hz / HIGH ~4-6 kHz; -infinity -> 0 -> +6 dB
Reroutes	3 other FX destinations per return; -infinity -> 0 dB; no self-route
Loops	Cross-FX feedback allowed

Element	Current specification
CUE	No FX CUE

- Active virtual-earth bus summing; nominal bus target ~0 dBu; earlier 10 k ohm injection / 10 k ohm feedback values are starting points only.
- Reroute path: return EQ -> VCA/attenuator -> low-Z driver -> injection resistor -> destination bus.
- Proposed reroute law: 0-20% -infinity->-40 dB; 20-45% -40->-20; 45-70% -20->-10; 70-85% -10->-4; 85-95% -4->-1; 95-100% -1->0 dB.
- Pedal pad ~-20 dB is prototype only. Return input target >=1 Mohm. Pedal interface target ~5 Hz-40 kHz.

10. Microphone and generic mono

Function	MIC	GENERIC MONO
Input	Balanced XLR	Mono jack
Gain	~30-60 dB	-infinity -> 0 -> +10 dB
Input Z	~2 k ohm	>=100 k ohm
EQ	LOW ~120 / HIGH ~5 kHz	LOW ~120 / HIGH ~5 kHz
Other	No CUE/filter/5-way/fader	No CUE/filter/5-way/fader

11. CUE / headphones

Function	Specification
CUE	DJ channels only; post-EQ/pre-filter/pre-fader
Headphone outputs	2
CUE/MASTER mix	CCW CUE / centre 50:50 / CW MASTER
Level	Shared
Other sources	FX/mic/ext excluded
Amp	Dedicated high-current isolated headphone amplifier

12. Metering

Colour	Meaning
BLUE	Little signal
GREEN	Safe
YELLOW	Loud
AMBER	Hard
RED	Extreme
PURPLE	Actual clipping

- DJ: post-EQ -> post-filter/drive -> pre-fader.
- 5-way: after drive, before master sum.
- Master: after master level.
- FX returns: after gain/EQ, pre-fader; proposed clip latch ~300-500 ms.
- Mic/generic: warning-only.

13. Power, grounding and backplane

Category	Current definition
Main rail	+/-24 V candidate
Analogue rail	+/-15 V candidate
Control	+5 / +3.3 V as required
Audio buses	DJ BUS, FX1-4, 5-WAY, MASTER, CUE all L/R
Grounds	Analogue 0 V, control ground, chassis/shield
Module rule	No internal-node access across modules
Hot swap	No
Test points	Every module and critical intermediate node

- Phono, mic and FX return circuitry must be physically separated from PSU, headphone and control-noise sources.

- Control CVs must be buffered and smoothed.
- Backplane impedance, loading, crosstalk and grounding remain open engineering tasks.

14. Noise, bandwidth and protection

Target	Current status
Critical-path noise	Proposed >= -90 dBu under defined reference conditions
Control->audio	Proposed >= -100 dB
Overall bandwidth	Proposed ~8 Hz-35 kHz with filters OFF
Protection	Required; not audio limiting
Power transitions	Output muting required

Correction: the -90 dBu complete-mixer figure and 8 Hz-35 kHz bandwidth remain targets to be calculated and measured, not assumed.

15. Major open engineering decisions

- Exact DJ EQ topology and values.
- Final RIAA schematic and cartridge-loading implementation.
- Global filter OTA and frequency-control architecture.
- Filter resonance/self-oscillation loop.
- Filter overdrive transfer curve.
- Five-way KILL filter-bank mathematics.
- Five-way creative EQ topology.
- FX VCA and control law.
- Pedal attenuation strategy.
- Master/booth/REC output topology.
- Power supply, grounding and EMC strategy.
- Backplane pinout, loading and isolation.
- Complete noise budget and PCB stack-up.

16. Recommended engineering build sequence

Step	Work package
1	One complete DJ channel, including the newly designed phono block
2	One FX send + pedal interface + return
3	Two FX buses with controlled feedback
4	One complete 5-way section
5	Master sum + output + booth/REC
6	CUE/headphones
7	Power/protection/muting
8	Backplane replication
9	Full measurement and integration

17. Reference calculations already discussed

- RIAA time constants: 3180 us, 318 us, 75 us.
- Simple RC reference: R = 1/(2 pi f C). With 10 nF: 70 Hz ~227 k ohm; 250 Hz ~63.7 k ohm; 1 kHz ~15.9 k ohm; 5 kHz ~3.18 k ohm. These are not final EQ values.
- A provisional 4.7 nF SVF integrator capacitor illustrated the 20 Hz-20 kHz control problem: estimated gm ~0.59 uS at 20 Hz and ~0.59 mS at 20 kHz.
- 10 k ohm injection / 10 k ohm feedback is a provisional active-summing starting point only.

18. Superseded / do not use

- Local per-channel filter frequency/resonance/overdrive controls - superseded.
- Mono final master - superseded; current architecture is stereo L/R.

- -20 dB pedal attenuation as a frozen universal value - prototype only.
- RIAA example RC values as a finished circuit - not authoritative.
- 0-5 V as mandatory native CV inside every circuit - not authoritative.
- +/-24 V as a universal op-amp supply - incorrect.
- Any provisional component value before topology selection - not frozen.

19. Phono design questions to close before Rev B

- Confirm that V1 is MM-only.
- Choose internal loading capacitance scheme: fixed 100 pF or technician-selectable 0/47/100/150 pF.
- Choose nominal 1 kHz gain: 40 dB or 42 dB. Current recommendation: about 42 dB.
- Choose whether loading should be technician-only or user-accessible on the rear panel.
- After these decisions, derive the complete RIAA schematic and simulate channel-to-channel tolerance, noise, overload and RIAA deviation before freezing component values.

20. Current bottom line

The project remains feasible. The phono section can now be treated as a concrete engineering work package rather than only a functional requirement. The current Rev A direction is MM-only, 47 k ohm load, 100 pF default with internal alternatives, JFET-input op-amp front end, approximately 21.3 dB first gain, hybrid active/passive RIAA, and approximately 40-42 dB total gain at 1 kHz.

Next engineering target: once the four phono questions are answered, freeze the actual RIAA topology and calculate the complete component network, then move to the DJ EQ or global filter.

21. Technical references consulted

- Texas Instruments, OPA1652/OPA1654 SoundPlus FET-input audio op-amp documentation. Relevant data: low input voltage/current noise, 18 MHz GBW, +/-18 V maximum supply range.
- Texas Instruments, AN-1651 / SNAA046A, Keeping Up with the Expanding Demands of High-Performance Audio. Documents passive and active/passive RIAA topologies and gain-distribution considerations.
- audioXpress, PhL-100 high-quality MM phono preamp article (2022/2023 web publication). Uses OPA1642, 21.3 dB first stage, passive 75 us section, second stage for 318 us and 3180 us, about 42 dB total gain, and reports measured RIAA accuracy.